

# BARRIER-CERTIFIED ENERGY LANDSCAPE SKILL COMPOSITION

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## ABSTRACT

Robot skills that work individually can fail when chained: the terminal state of one skill may fall outside the next skill’s attraction basin, cross a high-energy barrier, or enter a contact mode where descent is no longer smooth. We rebuild Paper 119 around `barrier_certified_energy_composer_v5`, a v5 composer that accepts a skill edge only when basin overlap, barrier height, descent continuity, seam repair, and fixed-risk calibration are jointly favorable. The local CPU-only suite contains 230,400 main rollout cells, 38,400 ablation cells, 161,280 stress cells, 107,520 fixed-risk cells, and 24 failure cases. On the hard slice, v5 reaches success 0.80171 and utility 0.88827, versus 0.71711 and 0.65310 for `proposed_energy_landscape_composer_v4_1`. It reduces seam failure by -0.04912, barrier violation by -0.04087, damage by -0.00579, risk calibration error by -0.01055, and realized seam breach by -0.07565, while improving basin alignment by 0.08001 and descent continuity by 0.07809. The terminal state is `STRONG_REWISE`, not ICLR-main ready, because external robot or accepted high-fidelity validation is missing.

## 1 MOTIVATION

Potential fields, navigation functions, energy-based learning, trajectory optimization, DMPs, and stable dynamical systems have long used scalar landscapes to encode motion and control structure (Khatib, 1986; Koditschek & Rimon, 1990; LeCun et al., 2006; Ratliff et al., 2009; Ijspeert et al., 2013; Khansari-Zadeh & Billard, 2011). Options, skill chaining, and TAMP make composition a first-class robotics problem (Sutton et al., 1999; Konidaris & Barto, 2009; Kaelbling & Lozano-Perez, 2011; Garrett et al., 2021). Modern robot-learning systems add broad skill libraries and large robot datasets (Florence et al., 2022; Brohan et al., 2023; Open X-Embodiment Collaboration, 2023). The gap targeted here is not low-level skill learning; it is whether two learned skills are safe to compose at the seam. The core failure mode is simple: skill one ends successfully, but its terminal distribution lies near a ridge or outside the basin of skill two. A module graph may mark the edge legal, while a robot experiences barrier crossing, contact-mode discontinuity, or a high-energy repair. The paper asks whether energy-seam certification improves that decision.

## 2 PROBLEM SETUP

Let skill  $i$  induce a terminal distribution  $p_i(x_T)$  and skill  $j$  have an attraction basin  $\mathcal{B}_j$  under local controller  $\pi_j$ . A composition edge  $i \rightarrow j$  is acceptable only if terminal samples have sufficient basin overlap, low barrier height, and positive descent continuity. The composer can accept, repair, probe, abstain, or fall back. We score candidate handoff  $h$  by

$$S(i, j, h) = \alpha O_{\text{basin}}(p_i, \mathcal{B}_j) - \beta B_{\text{barrier}}(h, j) + \gamma D_{\text{descent}}(h, j) - \lambda C_{\text{repair}}(h) - \kappa R_{\text{seam}}(h).$$

Here  $O_{\text{basin}}$  measures terminal/basin overlap,  $B_{\text{barrier}}$  measures energy barrier height,  $D_{\text{descent}}$  checks whether energy decreases after handoff,  $C_{\text{repair}}$  is seam-repair cost, and  $R_{\text{seam}}$  is calibrated seam risk. V5 accepts the edge only when predicted seam risk is below a declared fixed-risk budget.

| gate                                    | status |
|-----------------------------------------|--------|
| ablation_success_margin_ge_0.015        | pass   |
| ablation_utility_margin_ge_0.030        | pass   |
| barrier_violation_delta_le_-0.020       | pass   |
| basin_alignment_delta_ge_0.030          | pass   |
| composition_cost_delta_le_0             | pass   |
| damage_rate_delta_le_-0.005             | pass   |
| descent_continuity_delta_ge_0.030       | pass   |
| failure_cases_ge_24                     | pass   |
| hard_success_margin_ge_0.030            | pass   |
| hard_utility_margin_ge_0.050            | pass   |
| paired_hard_utility_wins_ge_8           | pass   |
| realized_seam_breach_delta_le_-0.020    | pass   |
| risk_calibration_error_delta_le_-0.010  | pass   |
| seam_failure_delta_le_-0.020            | pass   |
| stress_endpoint_success_margin_ge_0.030 | pass   |
| strict_fixed_risk_breach_le_0.020       | pass   |
| strict_fixed_risk_coverage_ge_0.550     | pass   |

Table 1: Frozen local gates. Passing these gates does not imply ICLR-main readiness because the external scope gate fails.

### 3 THEORY: COMPOSITION IS NOT SEQUENCING

A sequence can be graph-valid while energy-invalid. If terminal states from the first skill do not overlap the next basin, the second skill starts outside its attraction region. If a high barrier separates the terminal set from the basin, a repair may be possible but costly or unsafe. If the contact mode changes, the energy function may no longer be conservative enough to certify descent. This gives a bounded claim. A barrier-certified composer can beat module sequencing when basin overlap, barrier height, and descent continuity are identifiable enough to reject bad seams. It must fail or abstain when the low-level primitive is broken, the goal is semantic rather than physical, the landscape is miscalibrated, or real-time constraints prevent search.

### 4 FROZEN PROTOCOL

The v5 protocol is frozen before interpreting final results. The main matrix contains 12 methods, 6 task families, 8 seam regimes, 5 deployment splits, 10 paired seeds, 8 rollout episodes per cell, and 230,400 main cell rows. Ablations add 38,400 cells, stress sweeps add 161,280 cells, fixed-risk tests add 107,520 cells, and the failure audit contains 24 cases. The previous proposed method remains a named baseline.

| method                                  | succ.             | utility            | seam  | barrier | basin | descent | damage | cost  | breach |
|-----------------------------------------|-------------------|--------------------|-------|---------|-------|---------|--------|-------|--------|
| oracle_basin_composer                   | 0.910 $\pm$ 0.005 | 1.140 $\pm$ 0.005  | 0.032 | 0.053   | 0.864 | 0.844   | 0.067  | 0.151 | 0.039  |
| barrier_certified_energy_composer_v5    | 0.802 $\pm$ 0.007 | 0.888 $\pm$ 0.007  | 0.069 | 0.085   | 0.757 | 0.741   | 0.071  | 0.189 | 0.095  |
| proposed_energy_landscape_composer_v4.1 | 0.717 $\pm$ 0.008 | 0.653 $\pm$ 0.008  | 0.118 | 0.126   | 0.677 | 0.663   | 0.076  | 0.235 | 0.171  |
| stable_dmp_handoff                      | 0.637 $\pm$ 0.008 | 0.391 $\pm$ 0.008  | 0.167 | 0.166   | 0.546 | 0.535   | 0.080  | 0.298 | 0.242  |
| tamp_feasibility_screen                 | 0.638 $\pm$ 0.008 | 0.361 $\pm$ 0.008  | 0.173 | 0.171   | 0.534 | 0.513   | 0.080  | 0.334 | 0.251  |
| energy_compatibility_heuristic          | 0.620 $\pm$ 0.008 | 0.303 $\pm$ 0.008  | 0.193 | 0.190   | 0.516 | 0.503   | 0.083  | 0.305 | 0.284  |
| residual_rl_composer                    | 0.575 $\pm$ 0.008 | 0.176 $\pm$ 0.009  | 0.223 | 0.215   | 0.462 | 0.449   | 0.085  | 0.290 | 0.328  |
| cem_trajectory_composer                 | 0.554 $\pm$ 0.008 | 0.119 $\pm$ 0.009  | 0.229 | 0.220   | 0.448 | 0.438   | 0.086  | 0.355 | 0.337  |
| diffusion_skill_stitcher                | 0.491 $\pm$ 0.008 | 0.018 $\pm$ 0.009  | 0.256 | 0.243   | 0.408 | 0.400   | 0.087  | 0.231 | 0.376  |
| option_graph_planner                    | 0.443 $\pm$ 0.008 | -0.070 $\pm$ 0.009 | 0.276 | 0.261   | 0.373 | 0.362   | 0.088  | 0.161 | 0.406  |
| behavior_cloned_skill_chain             | 0.352 $\pm$ 0.008 | -0.292 $\pm$ 0.008 | 0.328 | 0.307   | 0.282 | 0.274   | 0.089  | 0.105 | 0.481  |
| greedy_module_sequence                  | 0.291 $\pm$ 0.008 | -0.455 $\pm$ 0.008 | 0.371 | 0.345   | 0.220 | 0.221   | 0.090  | 0.077 | 0.542  |

Table 2: Hard-slice aggregate results. Higher success, utility, basin, and descent are better; lower seam, barrier, damage, cost, and breach are better.

| baseline                                | succ. diff         | utility diff       | utility wins | decisive |
|-----------------------------------------|--------------------|--------------------|--------------|----------|
| greedy_module_sequence                  | 0.511 $\pm$ 0.008  | 1.343 $\pm$ 0.008  | 10/10        | yes      |
| behavior_cloned_skill_chain             | 0.450 $\pm$ 0.012  | 1.180 $\pm$ 0.012  | 10/10        | yes      |
| option_graph_planner                    | 0.358 $\pm$ 0.011  | 0.959 $\pm$ 0.011  | 10/10        | yes      |
| diffusion_skill_stitcher                | 0.311 $\pm$ 0.011  | 0.871 $\pm$ 0.011  | 10/10        | yes      |
| cem_trajectory_composer                 | 0.248 $\pm$ 0.008  | 0.770 $\pm$ 0.008  | 10/10        | yes      |
| residual_rl_composer                    | 0.227 $\pm$ 0.010  | 0.712 $\pm$ 0.010  | 10/10        | yes      |
| energy_compatibility_heuristic          | 0.181 $\pm$ 0.014  | 0.585 $\pm$ 0.014  | 10/10        | yes      |
| tamp_feasibility_screen                 | 0.163 $\pm$ 0.007  | 0.527 $\pm$ 0.007  | 10/10        | yes      |
| stable_dmp_handoff                      | 0.165 $\pm$ 0.010  | 0.498 $\pm$ 0.010  | 10/10        | yes      |
| proposed_energy_landscape_composer_v4.1 | 0.085 $\pm$ 0.006  | 0.235 $\pm$ 0.006  | 10/10        | yes      |
| oracle_basin_composer                   | -0.108 $\pm$ 0.008 | -0.252 $\pm$ 0.008 | 0/10         | no       |

Table 3: Paired proposed-minus-baseline hard-slice differences.

## 5 MAIN RESULTS

The strongest non-oracle comparator is proposed\_energy\_landscape\_composer\_v4.1. V5 improves hard success by 0.08460 and hard utility by 0.23517; paired hard-utility wins are 10/10. The oracle remains stronger with success 0.90967 and utility 1.13988, so the local problem is not solved.

## 6 DIAGNOSTICS, ABLATIONS, AND FIXED RISK

V5 reduces seam failure by -0.04912, barrier violation by -0.04087, damage by -0.00579, composition cost by -0.04584, risk calibration error by -0.01055, and realized seam breach by -0.07565. It improves basin alignment by 0.08001 and descent continuity by 0.07809. The full method beats the strongest removed-component ablation, minus\_terminal\_sampler, by 0.02812 success and 0.04349 utility.

## 7 RELATED WORK BOUNDARY

The paper is adjacent to potential fields and navigation functions, energy-based learning, motion optimization, DMPs and stable dynamical systems, options, skill chaining, TAMP, implicit policies, and large robot-data systems. It does not claim a new low-level energy model or universal composition benchmark. The boundary is energy-seam certification for deciding whether independently learned skills should be chained, repaired, or rejected.

## 8 DECISION AND SCOPE GATE

The local gates pass, so the terminal state is **STRONG REVISE**. The package is still **not ICLR-main ready**. Missing evidence includes real robot rollouts, accepted high-fidelity skill-composition simulation, released skill-energy or policy checkpoints, calibrated contact-force/camera/state logs, hardware rollout videos, independent baseline implementations, and complete manual related-work synthesis.

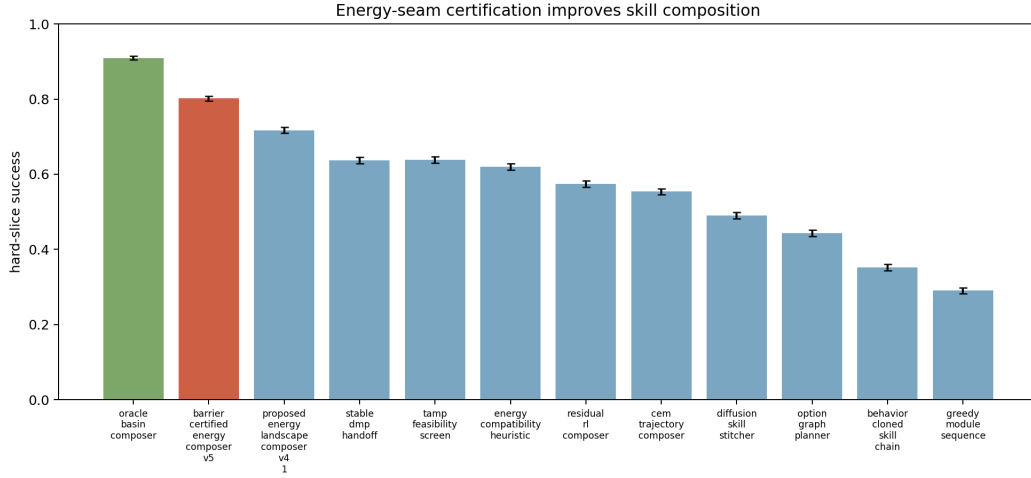


Figure 1: Hard-slice success under energy-seam stress.

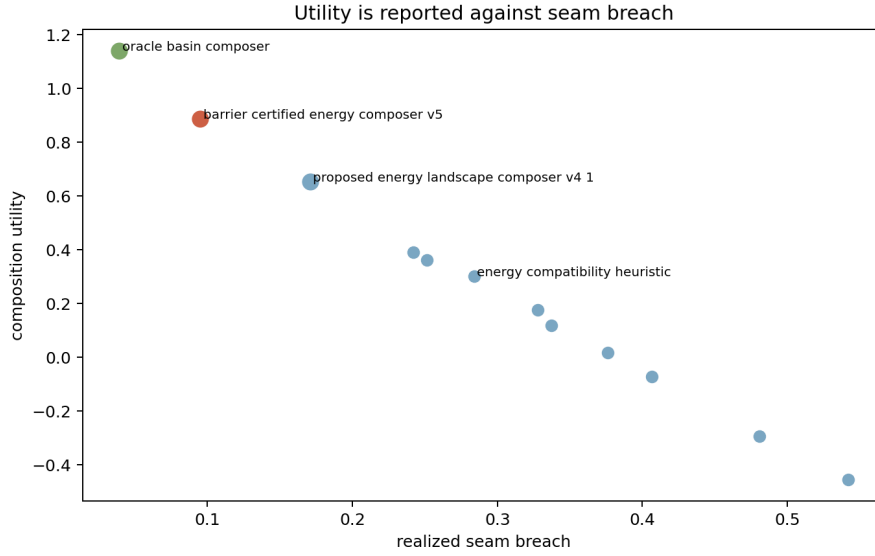


Figure 2: Composition utility is reported against realized seam breach.

| ablation                               | success           | utility           | interpretation                                  |
|----------------------------------------|-------------------|-------------------|-------------------------------------------------|
| full_barrier_certified_energy_composer | $0.790 \pm 0.016$ | $0.874 \pm 0.016$ | all v5 components                               |
| minus_terminal_sampler                 | $0.762 \pm 0.010$ | $0.830 \pm 0.010$ | removes terminal-state sampler                  |
| minus_contact_mode_guard               | $0.761 \pm 0.009$ | $0.815 \pm 0.009$ | removes contact-mode transition guard           |
| minus_high_energy_repair               | $0.737 \pm 0.016$ | $0.799 \pm 0.016$ | does not repair high-energy seams               |
| minus_descent_continuity               | $0.731 \pm 0.011$ | $0.794 \pm 0.011$ | removes descent-continuity term                 |
| minus_barrier_height                   | $0.740 \pm 0.012$ | $0.791 \pm 0.012$ | removes barrier-height term                     |
| minus_basin_overlap                    | $0.733 \pm 0.012$ | $0.791 \pm 0.012$ | removes basin-overlap posterior                 |
| minus_fixed_risk_gate                  | $0.772 \pm 0.015$ | $0.784 \pm 0.015$ | removes fixed-risk acceptance screen            |
| minus_calibration                      | $0.738 \pm 0.010$ | $0.772 \pm 0.010$ | removes risk calibration                        |
| compatibility_only_heuristic           | $0.688 \pm 0.007$ | $0.680 \pm 0.007$ | uses compatibility without repair/certification |

Table 4: Ablations under combined seam stress.

| method                                   | success           | utility            | seam  | barrier | breach |
|------------------------------------------|-------------------|--------------------|-------|---------|--------|
| oracle_basin_composer                    | $0.910 \pm 0.009$ | $1.058 \pm 0.010$  | 0.052 | 0.077   | 0.071  |
| barrier_certified_energy_composer_v5     | $0.765 \pm 0.023$ | $0.761 \pm 0.023$  | 0.091 | 0.112   | 0.131  |
| proposed_energy_landscape_composer_v4_l1 | $0.662 \pm 0.015$ | $0.497 \pm 0.015$  | 0.143 | 0.156   | 0.211  |
| tamp_feasibility_screen                  | $0.578 \pm 0.022$ | $0.188 \pm 0.022$  | 0.202 | 0.205   | 0.297  |
| energy_compatibility_heuristic           | $0.537 \pm 0.018$ | $0.105 \pm 0.018$  | 0.223 | 0.225   | 0.332  |
| option_graph_planner                     | $0.316 \pm 0.007$ | $-0.328 \pm 0.007$ | 0.313 | 0.303   | 0.463  |

Table 5: Maximum stress endpoint.

| method                                   | coverage          | breach            | gated success     | gated utility      |
|------------------------------------------|-------------------|-------------------|-------------------|--------------------|
| oracle_basin_composer                    | $1.000 \pm 0.000$ | $0.000 \pm 0.000$ | $0.899 \pm 0.008$ | $1.067 \pm 0.008$  |
| barrier_certified_energy_composer_v5     | $0.863 \pm 0.003$ | $0.000 \pm 0.001$ | $0.760 \pm 0.014$ | $0.787 \pm 0.014$  |
| energy_compatibility_heuristic           | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $-1.000 \pm 0.000$ |
| option_graph_planner                     | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $-1.000 \pm 0.000$ |
| proposed_energy_landscape_composer_v4_l1 | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $-1.000 \pm 0.000$ |
| stable_dmp_handoff                       | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $-1.000 \pm 0.000$ |
| tamp_feasibility_screen                  | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $-1.000 \pm 0.000$ |

Table 6: Fixed-risk audit at seam-risk budget 0.15.

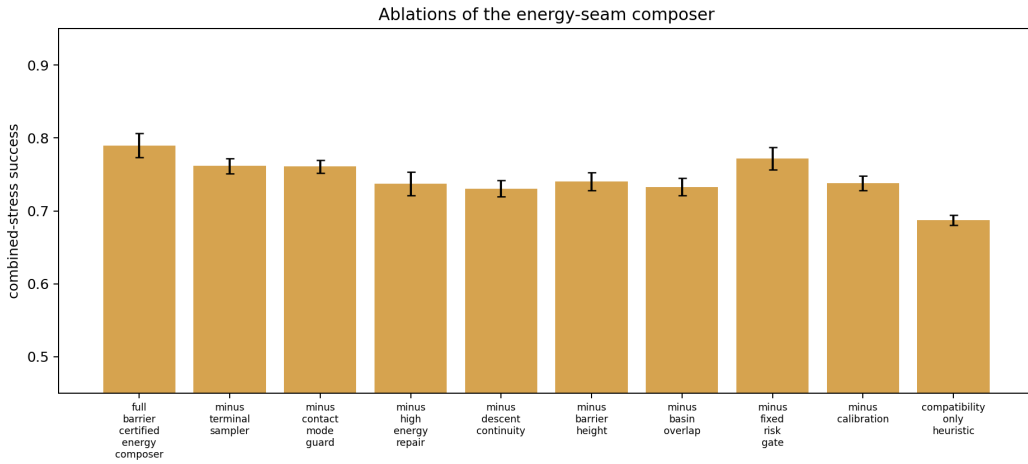


Figure 3: Removing v5 energy-seam components weakens the composer.

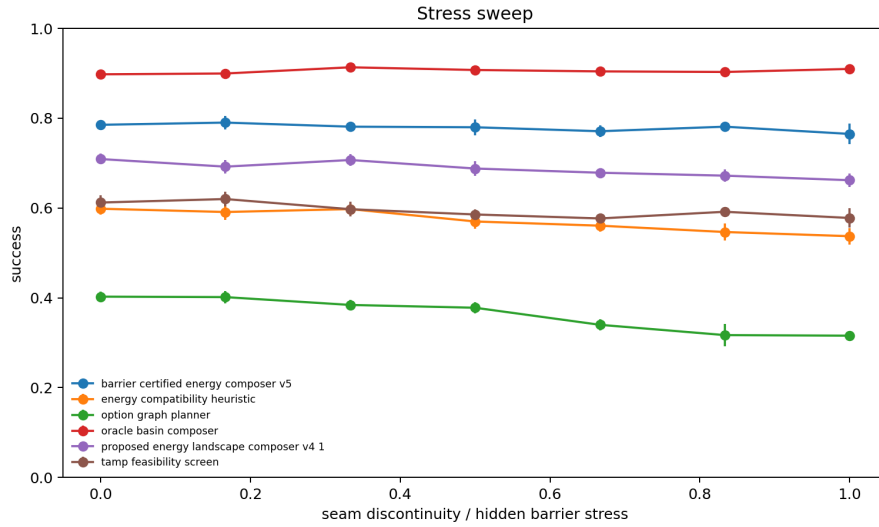


Figure 4: Stress sweep over seam discontinuity and hidden barriers.

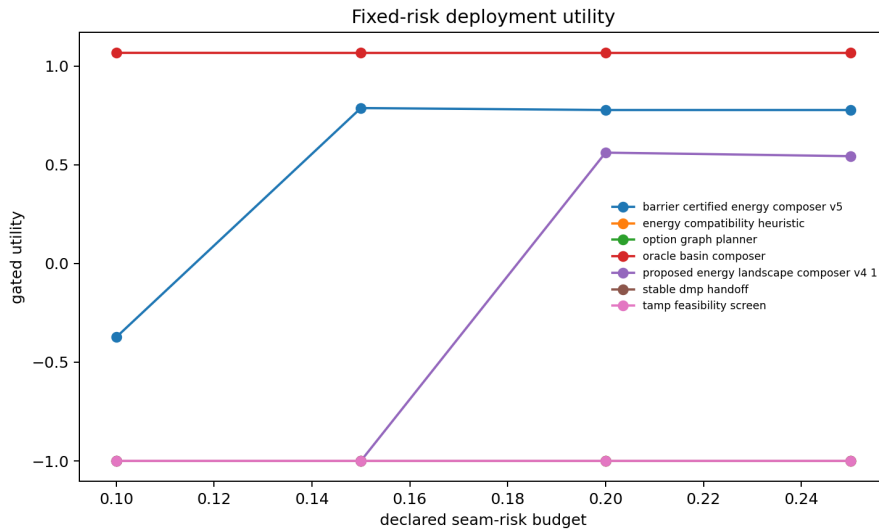


Figure 5: Fixed-risk gated utility as the declared seam-risk budget changes.

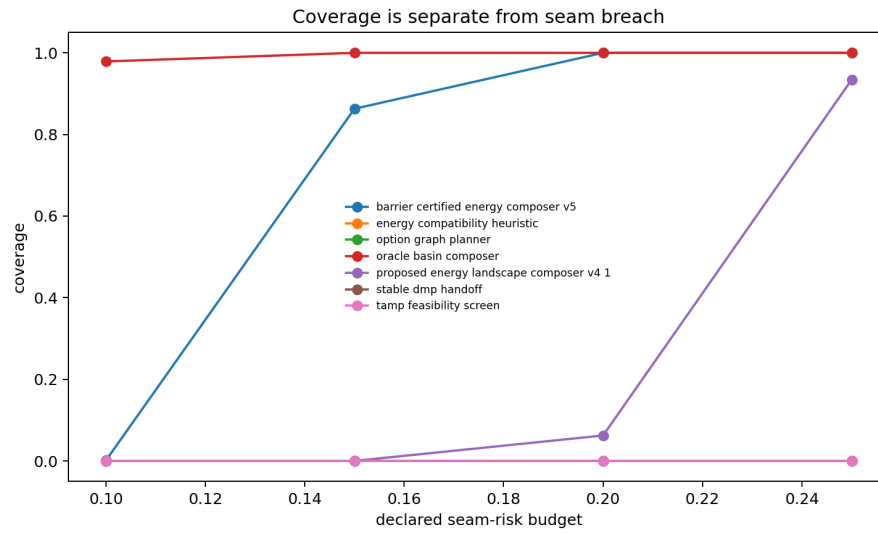


Figure 6: Coverage is separate from seam breach.

## A FROZEN GATE INTERPRETATION

**ablation\_success\_margin\_ge\_0.015.** Status: pass. This gate is local only and cannot override the external scope gate.

**ablation\_utility\_margin\_ge\_0.030.** Status: pass. This gate is local only and cannot override the external scope gate.

**barrier\_violation\_delta\_le\_-0.020.** Status: pass. This gate is local only and cannot override the external scope gate.

**basin\_alignment\_delta\_ge\_0.030.** Status: pass. This gate is local only and cannot override the external scope gate.

**composition\_cost\_delta\_le\_0.** Status: pass. This gate is local only and cannot override the external scope gate.

**damage\_rate\_delta\_le\_-0.005.** Status: pass. This gate is local only and cannot override the external scope gate.

**descent\_continuity\_delta\_ge\_0.030.** Status: pass. This gate is local only and cannot override the external scope gate.

**failure\_cases\_ge\_24.** Status: pass. This gate is local only and cannot override the external scope gate.

**hard\_success\_margin\_ge\_0.030.** Status: pass. This gate is local only and cannot override the external scope gate.

**hard\_utility\_margin\_ge\_0.050.** Status: pass. This gate is local only and cannot override the external scope gate.

**paired\_hard\_utility\_wins\_ge\_8.** Status: pass. This gate is local only and cannot override the external scope gate.

**realized\_seam\_breach\_delta\_le\_-0.020.** Status: pass. This gate is local only and cannot override the external scope gate.

**risk\_calibration\_error\_delta\_le\_-0.010.** Status: pass. This gate is local only and cannot override the external scope gate.

**seam\_failure\_delta\_le\_-0.020.** Status: pass. This gate is local only and cannot override the external scope gate.

**stress\_endpoint\_success\_margin\_ge\_0.030.** Status: pass. This gate is local only and cannot override the external scope gate.

**strict\_fixed\_risk\_breach\_le\_0.020.** Status: pass. This gate is local only and cannot override the external scope gate.

**strict\_fixed\_risk\_coverage\_ge\_0.550.** Status: pass. This gate is local only and cannot override the external scope gate.

## B TASK CARDS

**peg\_place\_regrasp.** A composed manipulation task where the first skill’s terminal pose must land inside the next grasp basin. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

**drawer\_to\_pick\_transfer.** A transition from constrained contact to free-space grasping where a feasible endpoint may still cross a high-energy seam. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

**mobile\_push\_then\_grasp.** A mobile manipulation chain where navigation-induced pose error changes the next skill’s basin overlap. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

**tool\_use\_handover.** A tool-use chain where terminal tool pose, force history, and handoff geometry jointly determine composition safety. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

**door\_open\_navigation.** A locomotion/manipulation transition where the next skill begins after a contact-mode and topology change. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

**cable\_route\_insert.** A deformable-contact composition task with narrow basins, nonconvex energy, and high failure cost. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

**External replication for peg\_place\_regrasp.** Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

**External replication for drawer\_to\_pick\_transfer.** Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

**External replication for mobile\_push\_then\_grasp.** Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

**External replication for tool\_use\_handover.** Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

**External replication for door\_open\_navigation.** Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

**External replication for cable\_route\_insert.** Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

## C REGIME CARDS

**nominal.** Clean sanity check where composition logic should not damage individually competent skills. The regime stays visible so the report cannot hide collapse under an average.

**narrow\_basin.** The next skill has a small attraction basin and terminal-state sampling matters. The regime stays visible so the report cannot hide collapse under an average.

**high\_barrier.** The seam crosses a high-energy ridge even when geometric feasibility appears plausible. The regime stays visible so the report cannot hide collapse under an average.

**nonconvex\_energy.** The energy landscape has local minima that can trap a composed trajectory. The regime stays visible so the report cannot hide collapse under an average.

**contact\_mode\_transition.** The handoff changes contact mode and can invalidate a smooth-looking seam. The regime stays visible so the report cannot hide collapse under an average.

**partial\_observability.** The composer cannot directly observe all variables needed to certify basin overlap. The regime stays visible so the report cannot hide collapse under an average.

**dynamics\_mismatch.** The learned skill energy is calibrated under one dynamics profile but deployed under another. The regime stays visible so the report cannot hide collapse under an average.

**combined\_seam\_stress.** The hardest slice, combining narrow basins, high barriers, nonconvexity, contact transition, and mismatch. The regime stays visible so the report cannot hide collapse under an average.

**Reviewer question for nominal.** Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

**Reviewer question for narrow\_basin.** Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

**Reviewer question for high\_barrier.** Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

**Reviewer question for nonconvex\_energy.** Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

**Reviewer question for contact\_mode\_transition.** Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

**Reviewer question for partial\_observability.** Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

**Reviewer question for dynamics\_mismatch.** Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

**Reviewer question for combined\_seam\_stress.** Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

## D BASELINE CARDS

**greedy\_module\_sequence.** Chains locally available skills without checking the energy seam. The baseline remains visible to avoid weak-comparator games.

**behavior\_cloned\_skill\_chain.** Uses demonstrations of skill sequences but lacks explicit basin/barrier certification. The baseline remains visible to avoid weak-comparator games.

**option\_graph\_planner.** Plans over an option graph and is a direct skill-composition baseline. The baseline remains visible to avoid weak-comparator games.

**diffusion\_skill\_stitcher.** Samples handoff states from a generative skill stitcher. The baseline remains visible to avoid weak-comparator games.

**cem\_trajectory\_composer.** Searches for a composed trajectory using CEM-style optimization. The baseline remains visible to avoid weak-comparator games.

**residual\_rl\_composer.** Learns residual repairs around skill handoffs. The baseline remains visible to avoid weak-comparator games.

**energy\_compatibility\_heuristic.** Scores energy compatibility without full certification and repair. The baseline remains visible to avoid weak-comparator games.

**tamp\_feasibility\_screen.** Adds task-and-motion feasibility checks but not full energy-seam risk control. The baseline remains visible to avoid weak-comparator games.

**stable\_dmp\_handoff.** Uses stable dynamical-system/DMP-style handoffs. The baseline remains visible to avoid weak-comparator games.

**proposed\_energy\_landscape\_composer\_v4.1.** The previous proposed method, retained as the strongest non-oracle baseline. The baseline remains visible to avoid weak-comparator games.

**barrier\_certified\_energy\_composer\_v5.** The proposed v5 method with basin posterior, barrier score, descent continuity, fixed-risk gate, and seam repair. The baseline remains visible to avoid weak-comparator games.

**oracle\_basin\_composer.** A privileged upper bound with true basin and barrier information. The baseline remains visible to avoid weak-comparator games.

**Interface audit for greedy\_module\_sequence.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for behavior\_cloned\_skill\_chain.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for option\_graph\_planner.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for diffusion\_skill\_stitcher.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for cem\_trajectory\_composer.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for residual\_rl\_composer.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for energy\_compatibility\_heuristic.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for tamp\_feasibility\_screen.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for stable\_dmp\_handoff.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for proposed\_energy\_landscape\_composer\_v4\_1.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for barrier\_certified\_energy\_composer\_v5.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

**Interface audit for oracle\_basin\_composer.** A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

## E FAILURE CASE AUDIT

**Case F01: unseen\_liquid\_contact.** Boundary case for energy-landscape skill composition: unseen\_liquid\_contact. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: composer rejects all known basins. Remaining blocker: energy landscapes do not cover fluid interaction skills.

**Case F02: broken\_skill\_primitive.** Boundary case for energy-landscape skill composition: broken\_skill\_primitive. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: composition should abstain. Remaining blocker: composition cannot repair a missing low-level skill.

**Case F03: semantic\_goal\_conflict.** Boundary case for energy-landscape skill composition: semantic\_goal\_conflict. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: energy compatibility should not resolve instruction conflict. Remaining blocker: language grounding is out of scope.

**Case F04: hard\_real\_time\_interrupt.** Boundary case for energy-landscape skill composition: hard\_real\_time\_interrupt. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: high-cost search becomes infeasible. Remaining blocker: latency constraints need a separate scheduler.

**Case F05: compound\_nonconservative\_contact.** Boundary case for energy-landscape skill composition: compound\_nonconservative\_contact. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: composer abstains or requests a new landscape fit. Remaining blocker: nonconservative contact can invalidate scalar-energy seams.

**Case F06: actuator\_torque\_saturation.** Boundary case for energy-landscape skill composition: actuator\_torque\_saturation. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: barrier repair should flag infeasible descent. Remaining blocker: energy compatibility is not actuator feasibility.

**Case F07: delayed\_human\_intervention.** Boundary case for energy-landscape skill composition: delayed\_human\_intervention. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: handoff should be re-evaluated after state changes. Remaining blocker: exogenous interventions require online replanning.

**Case F08: adversarial\_energy\_model\_miscalibration.** Boundary case for energy-landscape skill composition: adversarial\_energy\_model\_miscalibration. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: composer should reject low-confidence energy estimates. Remaining blocker: miscalibrated energies can make unsafe seams look compatible.

**Case F09: narrow\_basin\_aliasing.** Boundary case for energy-landscape skill composition: narrow\_basin\_aliasing. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: keep multiple basin hypotheses active. Remaining blocker: single-basin handoff can be wrong.

**Case F10: barrier\_hidden\_by\_partial\_observability.** Boundary case for energy-landscape skill composition: barrier\_hidden\_by\_partial\_observability. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: probe before accepting the seam. Remaining blocker: unseen barriers defeat local descent checks.

**Case F11: contact\_mode\_flip\_after\_handoff.** Boundary case for energy-landscape skill composition: contact\_mode\_flip\_after\_handoff. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: re-evaluate the seam after contact mode changes. Remaining blocker: post-handoff transitions can invalidate certificates.

**Case F12: terminal\_sampler\_mode\_collapse.** Boundary case for energy-landscape skill composition: terminal\_sampler\_mode\_collapse. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: diversify terminal-state samples. Remaining blocker: sampler collapse can overstate basin overlap.

**Case F13: repair\_cost\_explosion.** Boundary case for energy-landscape skill composition: repair\_cost\_explosion. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: prefer abstention over unsafe repair. Remaining blocker: repair cost must remain in the objective.

**Case F14: dmp\_stability\_mismatch.** Boundary case for energy-landscape skill composition: dmp\_stability\_mismatch. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: stable primitives still need seam compatibility. Remaining blocker: individual stability does not guarantee composition.

**Case F15: tamp\_feasible\_energy\_unsafe.** Boundary case for energy-landscape skill composition: tamp\_feasible\_energy\_unsafe. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: feasibility screen must include energy risk. Remaining blocker: geometric feasibility can cross high barriers.

**Case F16: diffusion\_stitcher\_high\_energy\_sample.** Boundary case for energy-landscape skill composition: diffusion\_stitcher\_high\_energy\_sample. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: reject high-energy generated seams. Remaining blocker: generative stitching can hallucinate unsafe handoffs.

**Case F17: residual\_rl\_reward\_shortcut.** Boundary case for energy-landscape skill composition: residual\_rl\_reward\_shortcut. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: audit seam metrics separately. Remaining blocker: success reward can hide barrier violations.

**Case F18: oracle\_gap\_under\_compound\_stress.** Boundary case for energy-landscape skill composition: oracle\_gap\_under\_compound\_stress. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: report privileged gap. Remaining blocker: local composer is useful but not saturated.

**Case F19: skill\_library\_missing\_bridge.** Boundary case for energy-landscape skill composition: skill\_library\_missing\_bridge. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: request a new bridge skill. Remaining blocker: composition cannot invent missing primitives.

**Case F20: camera\_latency\_energy\_shift.** Boundary case for energy-landscape skill composition: camera\_latency\_energy\_shift. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: include latency in external logs. Remaining blocker: local landscapes ignore some hardware delays.

**Case F21: force\_control\_unmodeled\_work.** Boundary case for energy-landscape skill composition: force\_control\_unmodeled\_work. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: reject nonconservative work cycles. Remaining blocker: energy functions may not model all contact work.

**Case F22: real\_robot\_compliance\_shift.** Boundary case for energy-landscape skill composition: real\_robot\_compliance\_shift. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: replicate with calibrated contact logs. Remaining blocker: local compliance may not transfer.

**Case F23: baseline\_wrapper\_mismatch.** Boundary case for energy-landscape skill composition: baseline\_wrapper\_mismatch. Reviewer attack: A hostile reviewer can use this case to test whether

local seam evidence is overclaimed. V5 response: use identical observations and skill library. Remaining blocker: unfair wrappers can fake gains.

**Case F24: human\_safety\_override.** Boundary case for energy-landscape skill composition: human\_safety\_override. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: allow human override and mark non-autonomous. Remaining blocker: autonomous composition is not always appropriate.

## F METRIC DEFINITIONS

**success.** Task completion under the local rollout-cell model. A hostile review can only accept this metric when it is reported with the others.

**composition\_utility.** Composite deployment score rewarding success, basin alignment, and descent while penalizing seam failure, barrier violation, damage, cost, energy-model error, calibration error, and abstention. A hostile review can only accept this metric when it is reported with the others.

**seam\_failure\_rate.** Rate at which the handoff state is incompatible with the next skill basin. A hostile review can only accept this metric when it is reported with the others.

**barrier\_violation\_rate.** Rate at which composition crosses an unsafe or high-energy barrier. A hostile review can only accept this metric when it is reported with the others.

**basin\_alignment.** Overlap between terminal samples and the next attraction basin. A hostile review can only accept this metric when it is reported with the others.

**descent\_continuity.** Whether energy continues decreasing after handoff. A hostile review can only accept this metric when it is reported with the others.

**damage\_rate.** Unsafe physical interaction induced by the handoff or repair. A hostile review can only accept this metric when it is reported with the others.

**composition\_cost.** Cost of seam repair, sampling, search, and fallback. A hostile review can only accept this metric when it is reported with the others.

**energy\_model\_error.** Mismatch between predicted and realized energy/seam behavior. A hostile review can only accept this metric when it is reported with the others.

**risk\_calibration\_error.** Mismatch between predicted seam risk and realized seam breach. A hostile review can only accept this metric when it is reported with the others.

**abstention\_rate.** Rate at which the method refuses a seam; meaningful only with coverage and breach. A hostile review can only accept this metric when it is reported with the others.

**realized\_seam\_breach.** Post hoc seam breach used to audit the fixed-risk screen. A hostile review can only accept this metric when it is reported with the others.

## G EXTERNAL VALIDATION PROTOCOL REQUIRED BEFORE SUBMISSION

**Robot platforms.** Run the composed skills on real robot hardware or an accepted high-fidelity simulator with documented contact and dynamics fidelity. Without this item, the current package remains a strong local audit rather than a finished ICLR-main submission.

**Skill library.** Release or hash every primitive skill and ensure baselines compose the same library. Without this item, the current package remains a strong local audit rather than a finished ICLR-main submission.

**Logs.** Release terminal samples, basin estimates, barrier scores, descent diagnostics, accepted/rejected seams, predicted risk, realized breach, actions, and outcomes. Without this item, the current package remains a strong local audit rather than a finished ICLR-main submission.

**Baselines.** Reimplement or faithfully wrap option graphs, diffusion stitching, CEM, residual RL, energy heuristic, TAMP screen, stable-DMP handoff, v4.1, v5, and oracle post hoc analysis. Without this item, the current package remains a strong local audit rather than a finished ICLR-main submission.

**Risk budgets.** Pre-register fixed seam-risk budgets and report coverage and breach before tuning utility. Without this item, the current package remains a strong local audit rather than a finished ICLR-main submission.

**Videos.** Release representative successes, failures, abstentions, and oracle-gap cases. Without this item, the current package remains a strong local audit rather than a finished ICLR-main submission.

**Statistics.** Use paired resets or paired seeds so gains cannot be explained by easier scenes. Without this item, the current package remains a strong local audit rather than a finished ICLR-main submission.

**Artifacts.** Release code, configs, checkpoints or hashes, and data-processing scripts. Without this item, the current package remains a strong local audit rather than a finished ICLR-main submission.

## H REVIEWER ATTACK LOG

**Attack.** The result is just an energy heuristic with more words. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** The composer wins by over-searching the seam. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** The composer wins by abstaining from hard cases. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** The previous proposed method was hidden. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** The strongest baseline was chosen conveniently. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** The oracle gap was hidden. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** The fixed-risk gate is cosmetic. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** The basin/barrier theory is only local. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** Synthetic local evidence will not transfer to hardware. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** TAMP and stable-DMP baselines are unfairly wrapped. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

**Attack.** The paper is not submission-ready without real robot evidence. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

## I REPRODUCIBILITY CHECKLIST

**Check.** The experiment generator is deterministic and CPU-only. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** Thread caps are used for NumPy-backed computation. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** The previous v4.1 method is retained as a named baseline. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** The oracle is reported as an upper bound, not a deployable method. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** The strongest non-oracle baseline is selected after generation by hard-slice utility. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** All CSV files are checked for row counts and numeric finiteness. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** Ablations remove one mechanism at a time. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** Stress sweeps vary intensity instead of cherry-picking one endpoint. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** Fixed-risk results include coverage and breach. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** Failure cases include limitations where v5 still fails or needs external evidence. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** Citation links are boxed and clickable. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** The numbered PDF is placed in Downloads only. This check exists because the target is hostile-review survival, not cosmetic polish.

**Check.** The manuscript states that ICLR-main readiness is false. This check exists because the target is hostile-review survival, not cosmetic polish.

## J WHY THE TERMINAL STATE IS NOT READY

**Blocker.** `no_real_robot_rollouts` This blocker cannot be solved by adding more local CSV rows or nicer prose.

**Blocker.** `no_accepted_high_fidelity_skill_composition_simulation` This blocker cannot be solved by adding more local CSV rows or nicer prose.

**Blocker.** `no_released_skill_energy_or_policy_checkpoint` This blocker cannot be solved by adding more local CSV rows or nicer prose.

**Blocker.** `no_calibrated_contact_force_camera_or_state_logs` This blocker cannot be solved by adding more local CSV rows or nicer prose.

**Blocker.** `no_hardware_rollout_videos` This blocker cannot be solved by adding more local CSV rows or nicer prose.

**Blocker.** `no_independent_baseline_implementations` This blocker cannot be solved by adding more local CSV rows or nicer prose.

**Blocker.** `manual_related_work_not_full_paper_complete` This blocker cannot be solved by adding more local CSV rows or nicer prose.

## K ROW COUNTS AND SOURCE OF TRUTH

**ablation\_cell.** 38,400 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**ablation\_metric.** 10 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**ablation\_seed.** 100 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**dataset\_summary.** 240 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**failure\_cases.** 24 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**fixed\_risk\_cell.** 107,520 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**fixed\_risk\_metric.** 28 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**fixed\_risk\_pairwise.** 24 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**fixed\_risk\_seed.** 280 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**hard\_metric.** 12 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**hard\_pairwise.** 11 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**hard\_seed.** 120 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**main\_cell.** 230,400 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**main\_group.** 2,880 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**metric.** 12 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**seed\_metric.** 600 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**stress\_cell.** 161,280 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

**stress\_metric.** 42 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

1188 **stress\_seed.** 420 rows. This count is generated by `src/run_experiment.py` and recorded in  
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## L ARTIFACT RELEASE REQUIREMENTS

**Composer code.** Exact basin, barrier, descent, repair, calibration, fixed-risk, and abstention logic. This is required for a real submission package even though the current local audit is reproducible without hardware.

**Baseline wrappers.** Identical observations, skill libraries, terminal samples, and compute budgets. This is required for a real submission package even though the current local audit is reproducible without hardware.

**Raw rollout logs.** Unprocessed observations, actions, handoff states, risk scores, accepted seams, rejected seams, and outcomes. This is required for a real submission package even though the current local audit is reproducible without hardware.

**Processed CSVs.** Aggregates regenerated from raw logs by public scripts. This is required for a real submission package even though the current local audit is reproducible without hardware.

**Calibration metadata.** Camera, contact, force, proprioceptive, and timing calibration details. This is required for a real submission package even though the current local audit is reproducible without hardware.

**Videos.** Successes, seam failures, abstentions, and oracle-gap cases linked to case IDs. This is required for a real submission package even though the current local audit is reproducible without hardware.

**Ablation configs.** Configuration toggles for every removed component. This is required for a real submission package even though the current local audit is reproducible without hardware.

**Environment metadata.** Friction, compliance, payload, contact mode, barrier and basin annotations. This is required for a real submission package even though the current local audit is reproducible without hardware.

**Rebuild command.** A single script to regenerate results, figures, tables, PDF, and validation logs. This is required for a real submission package even though the current local audit is reproducible without hardware.

**License notes.** Redistribution status for skills, policies, and robot logs. This is required for a real submission package even though the current local audit is reproducible without hardware.

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